

Touchpoint Analytics: Technical Business Case & Architecture Review

Comprehensive End-to-End Implementation Guide

Chapter 1: The Corporate Profile & Case Study Alignment

THE BEVERAMA CORPORATE ESSENCE:

To build a truly impactful Business Intelligence solution, we must align the data architecture with the company's DNA as outlined in the Case Study:

1. Global Market Leader: Beverama is the world's largest beverage producer with a 33% global market share and 57 locations worldwide. The UK operations are based in Scarborough.
2. Brand Ethos ('It's all tasty'): The company focuses on the quality of its beverages and strong relationships with suppliers, communities, and customers.
3. Sustainable Growth: Beverama invests heavily in alternative energy (e.g., wind turbines) and supports local charities and social impact projects.
4. Product Portfolio & Innovation: The portfolio covers all consumer trends: Beverama (natural juices), Bevty (anti-oxidant blends), Juiced (affordable range), and Wipe Bevs (fortified blends). The focus is on healthier drinks and adapting to evolving lifestyles.
5. Local Market Strategy: This dataset represents a recent entry into the local market, focusing specifically on the two largest retail chains (medium and large-format supermarkets).

By grounding our BI solution in this corporate profile, we ensure that every metric and insight serves the ultimate goal: strengthening Beverama's position in this new local market while respecting its global ethos.

Chapter 2: The Business Story & Time Window (The Pitch)

THE PITCH (Memorize this for the interview):

"I analyzed a comprehensive dataset spanning a Time Window of exactly 25 months (from January 1, 2018, to January 27, 2020). The goal was to dissect EUR 2.7M in total revenue to understand exactly 'what happened' and 'how it happened'.

WHAT HAPPENED:

Over these two years, the business exhibited extreme seasonality. We observe massive revenue spikes at the end of each year, followed by sharp contractions in the first quarter. Despite these peaks, the overarching trend required deeper investigation into product pricing and geographic concentration.

HOW IT HAPPENED (The Sub-Brand Portfolio Strategy):

1. The Parent Company vs Sub-Brand Dynamic: As per the case study, 'Beverama' is the parent company, but its namesake natural juice line ('Beverama') is the #1 anchor of the sub-brand portfolio, driving the highest volume and revenue (EUR 720k).
2. The Core Pillars (Juiced & Wipe Bevs): These are not niche sub-brands; they are massive financial pillars, generating over EUR 600k each and moving tremendous volume, rivaling the leader.

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3. The Premium Niche (Bevty): Operates at a higher price point (Premium), driving EUR 380k in revenue despite having only a fraction of the volume of the others.
4. Geographic Concentration: Cities like Detroit and Minneapolis act as heavy financial anchors, dominating the client footprint.
5. The 'Unknown SKU' Strategy: Over 110k units sold were unmapped to any Master Data product. A junior analyst would drop these rows, losing critical financial data. Instead, I strategically aggregated them under the parent 'Beverama' brand. This preserves 100% of global Revenue for the C-level, while keeping the 'Unknown SKU' flag at the product level for the MDM team to urgently fix.

With this historical context established, I built an AI predictive model to forecast the next 6 months and engineered a dynamic dashboard so managers can slice this story by any sub-brand, city, or date."

Chapter 3: Executive Strategy & Architecture Selection

This project was designed from the ground up to demonstrate a 'Senior Data Engineer' approach rather than a junior script. Instead of relying on standard Pandas (which is notorious for Out-Of-Memory errors on production-scale data), the pipeline leverages DuckDB.

Architecture Choice: Native ELT (Extract, Load, Transform) via Medallion Architecture.

- Bronze Layer: Raw CSV files are read natively into memory without strict typing.
- Silver Layer: Pure SQL is used for cleansing, type casting, and anomaly removal.
- Gold Layer: Data is normalized into a strict Kimball Star Schema.

This approach proves that the candidate understands how to scale analytics to gigabytes of data using columnar, in-process engines, minimizing computational overhead on the final Streamlit dashboard.

Chapter 4: Data Extraction & Cleansing (Silver Layer)

The raw data contained several format inconsistencies and mathematical anomalies that required handling:

1. String Cleansing: The 'PVP' (Price) column contained 'EUR' currency symbols and European comma separators. These were surgically removed via SQL REPLACE functions and CAST to DOUBLE.
2. Upstream Pre-calculation: Revenue was calculated directly in the ELT layer ($\text{Price} * \text{Volume}$) rather than forcing the Streamlit dashboard to calculate it on the fly. This saves critical CPU cycles during UI rendering.
3. Anomaly Expurgation: An extreme logic audit revealed 16 transactions with zero or negative Volume/Revenue (likely customer returns). A strict WHERE clause was added to the Silver layer to drop these records, ensuring that downstream metrics like 'Average Price' ($\text{Revenue} / \text{Volume}$) are mathematically safe from division-by-zero errors.

Chapter 5: Kimball Star Schema & Data Quality (Gold Layer)

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The core of the Business Intelligence engine is a robust Relational Model adhering to Ralph Kimball's rules.

- Dim_Store (1:1 Granularity): The 'Store' and 'Client' raw files were joined. A critical duplication was solved by recognizing that 'Cliente' and 'BannerID' together form a composite key, ensuring perfect uniqueness.
- Dim_Date: A programmatic continuous calendar was generated from Jan 2018 to Dec 2020. This allows for flawless Time-Intelligence.
- Dim_Item & The 'Unknown SKU' Strategy: The raw Fact table contained 11 ITEMCODEs that did not exist in the Master Items list. A junior analyst would drop these rows (losing financial data) or fail the pipeline. Instead, the ETL dynamically maps missing codes to an 'Unknown SKU' dimension. This brilliant strategy preserves 100% of global Revenue while mathematically flagging Master Data inconsistencies for the business to fix.

Chapter 6: Dashboard Logic & Mathematical Edge Cases

The Streamlit application was built to be highly interactive, but filters can break simple math. Several advanced protections were coded:

1. Year-to-Date (YTD) YoY Alignment: The dataset ends abruptly on January 27, 2020. A naive YoY calculation would compare 27 days of 2020 against 365 days of 2019, resulting in a false -95% drop. The code enforces a strict YTD limit on the previous year matching the exact day-of-year of the current year.
2. The Target Gauge: Instead of arbitrary fixed numbers, the KPI Target Gauge calculates the 'All-Time Best Historic Year' for the selected filters. This dynamic benchmark constantly challenges executives to beat their best historical performance.

Chapter 7: KPI Strategy & Metric Definitions (Business Justifications)

Every metric on the dashboard was carefully selected to answer specific executive questions:

1. Total Revenue: The ultimate anchor of business health. Tells C-Levels exactly how much money the business generated under the selected filters.
2. Total Volume (Units): Revenue can grow artificially through price inflation while market share shrinks. Tracking Volume ensures we monitor actual product demand.
3. Average Price / Unit (Revenue / Volume): A crucial profitability metric. By tracking this, managers can instantly see if discounting strategies are hurting margins or if premium products are driving growth.
4. YoY Growth (Year-over-Year): Calculated dynamically with strict Year-to-Date alignment. Justification: Absolute numbers lack context. A manager needs to know the 'momentum' of the business (growing or shrinking compared to the same period last year).
5. Best Year Benchmark (Target Gauge): Instead of arbitrary static targets, the Gauge calculates the 'All-Time Historic High' for the exact filter context. This visually challenges managers to constantly beat their own historical records.
6. Machine Learning Forecast (6-Months): Descriptive KPIs only tell what happened. The AI Forecast provides a mathematical projection of where the Revenue is heading, allowing Supply Chain and Finance to prepare in advance.

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Chapter 8: Data Visualization & Best Practices

The UI follows 2026 Executive Dashboard aesthetics:

- Sparkline Indicators: Standard KPIs were upgraded to Plotly Sparklines. Numbers are overlayed on historical trend areas, providing instant context.
- Waterfall Chart: Bridges the exact category-level drivers behind Revenue growth or contraction between years.
- Pareto Chart (80/20 Rule): A cumulative percentage line graph instantly reveals which core Brands drive the vast majority of the company's financial weight.
- Scatter Plots: By plotting Price vs Volume, Product Managers can instantly visualize market positioning and identify 'Cash Cows'.

Chapter 9: Predictive Analytics (XGBoost) & ML Poisoning

To move beyond Descriptive Analytics into Predictive Intelligence, an XGBoost Regression model was implemented.

- Feature Engineering: The model learns from Time Indexes, Lag_1 (previous month), Lag_2, and a 3-month Rolling Mean.
- Preventing ML Poisoning: Since Jan 2020 only has 27 days, its total volume is artificially low. Training the model on this incomplete month would 'poison' the AI, causing it to predict a massive future contraction. The pipeline detects incomplete final months and gracefully excludes them from the training set, ensuring the forecast remains highly accurate and stable.

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Chapter 10: Enterprise Scalability Roadmap (Phase 2)

While perfectly tailored for this case study, scaling this architecture to 100 Million+ rows requires the following Enterprise upgrades:

1. Schema Contracts: Replace 'read_csv_auto' with explicit schema definitions to prevent pipeline crashes if upstream source data types drift.
2. Currency Agnosticism: Replace hardcoded 'EUR' replacement with Regex (Regular Expressions) to extract numerics, supporting future global expansion.
3. Incremental Upserting: Move away from 'Full Load' (CREATE OR REPLACE) to 'Incremental Load', checking the MAX(Date) and importing only net-new daily data.
4. ML Caching & Tuning: Implement '@st.cache_resource' to prevent the model from retraining on every UI click, and add GridSearch for dynamic Hyperparameter Tuning.